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A symmetry on parking functions via Dyck paths

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ABSTRACT

This paper proves a symmetric joint distribution on parking functions via constructing two involutions on Dyck paths. The second involution is built with the aid of plane trees, leading to the discovery of two natural symmetric pairs on rooted labeled trees.

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1. Introduction

Let $[n] := \{1, 2, \ldots, n\}$. A sequence $\alpha = (a_1, a_2, \ldots, a_n) \in [n]^n$ is called a *parking function* if its non-decreasing rearrangement $\mathrm{inc}(\alpha) = (b_1, b_2, \ldots, b_n)$ satisfies $b_i \leqslant i$ for every i. Consider a one-way street with ordered parking spaces numbered $1, 2, \ldots, n$ and interpret a_i as the preferred parking space of the car V_i $(i = 1, 2, \ldots, n)$, then α is a parking function if and only if at least i cars prefer to park at the first i spaces, or equivalently, each car can park in this street using the rules below:

- A car tries to park in its preferred space, but if that space is occupied, then it parks in the next available space.
- If there is no available space after its preferred space then the car leaves the street.

The notion of parking functions was introduced by Konheim and Weiss [7] in their study of computer storage. Using analytic methods, they proved that $|\mathbf{PF}_n| = (n+1)^{n-1}$ where $\mathbf{PF}_n$ denotes the set of parking functions of length n. A simple and elegant proof was provided by Pollak (see [9]). Parking functions are related to labeled trees, hyperplane arrangements, noncrossing partitions and Dyck paths; see the excellent survey of Yan [9] for further information.

A parking function is said to be *prime* if removing one entry equal to 1 from it still remains a parking function. Equivalently, a parking function $(a_1, a_2, \ldots, a_n)$ is prime if $|\{i \in [n] : a_i \leqslant j\}| \geqslant j + 1$ for all $1 \leqslant j \leqslant n - 1$. Prime parking functions and minimal transitive cycle factorizations of permutations are related very closely [6]. The notion of prime parking functions was introduced by Gessel [8, p. 141], who observed that any parking function can be decomposed into some prime parking functions. It is more convenient to describe Gessel's decomposition of parking functions reversely. Given a sequence $\alpha_1, \ldots, \alpha_k$ of prime parking functions such that α_i has length d_i, form β_i by adding $d_1 + d_2 + \cdots + d_{i-1}$ to all entries of α_i. Then any shuffle of all the β_i's is a parking function, and conversely given any parking function one can uniquely recon-

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struct $\alpha_1, \ldots, \alpha_k$. For example, the parking function 167426148 is decomposed into three prime parking functions 121, 11 and 1213.

It then follows from Gessel's decomposition and the product formula [8, Prop. 5.1.3] for exponential generating functions that

$$\sum_{n\geqslant 0}(n+1)^{n-1}\frac{x^n}{n!}=\left(1-\sum_{n\geqslant 1}P(n)\frac{x^n}{n!}\right)^{-1}, \tag{1.1}$$

where $P(n)$ is the number of prime parking functions of length n. Comparing with [8, Eq. (5.67)], an identity derived from the Lagrange inversion formula, we conclude the following well-known formula

$$P(n)=(n-1)^{n-1}. \tag{1.2}$$

A cycle lemma approach to (1.2) was provided by L. Kalikow (see [8, p. 141]). Moreover, if we denote $\mathsf{pri}(\alpha)$ the number of prime parking functions decomposed from α, then by Gessel's decomposition and Eq. (1.2) we have the following refinement of (1.1)

$$\sum_{n\geqslant 0}\sum_{\alpha\in\mathbf{PF}_n}t^{\mathsf{pri}(\alpha)}\frac{x^n}{n!}=\left(1-t\sum_{n\geqslant 1}(n-1)^{n-1}\frac{x^n}{n!}\right)^{-1}. \tag{1.3}$$

On the other hand, motivated by their study of the Pak–Stanley labeling of chambers of Shi arrangement, Duarte and Guedes de Oliveira [4] introduced two statistics, called *runs* and *centers*, on parking functions that turn out to have the same distribution as "pri". For $\alpha=(a_1,\ldots,a_n)\in\mathbf{PF}_n$, define respectively the run of α and the size of the center of α by

- $\mathsf{run}(\alpha)=\max\{k\in[n]:[k]\subseteq\{a_1,a_2,\ldots,a_n\}\}$
- and $\mathsf{z}(\alpha)=\max\{k\in[n]:\exists\, i_1<\cdots<i_k$ such that $a_{i_1}\leqslant 1,\ldots,a_{i_k}\leqslant k\}$.

Then, there holds the following equidistribution

$$\sum_{\alpha\in\mathbf{PF}_n}t^{\mathsf{z}(\alpha)}=\sum_{\alpha\in\mathbf{PF}_n}t^{\mathsf{run}(\alpha)}=\sum_{\alpha\in\mathbf{PF}_n}t^{\mathsf{pri}(\alpha)},$$

where the first equality was established in [4], while the second equality was proved only recently by Du [3]. The main purpose of this paper is to prove the symmetry of the joint distribution $(\mathsf{run},\mathsf{pri})$ on parking functions using two different involutions on Dyck paths.

Theorem 1.1. *For $n\geqslant 1$, we have the symmetry*

$$\sum_{\alpha\in\mathbf{PF}_n}s^{\mathsf{pri}(\alpha)}t^{\mathsf{run}(\alpha)}=\sum_{\alpha\in\mathbf{PF}_n}s^{\mathsf{run}(\alpha)}t^{\mathsf{pri}(\alpha)}. \tag{1.4}$$

Remark 1.2. Note that neither the pair $(\mathsf{z},\mathsf{run})$ nor the pair $(\mathsf{z},\mathsf{pri})$ is symmetric on parking functions.

It is well known that rooted labeled trees on $\{0,1,\ldots,n\}$ with root 0 are also enumerated by $(n+1)^{n-1}$. In view of Theorem 1.1, one may wonder whether there exists any pair of statistics over rooted labeled trees that has the same distribution as $(\mathsf{run},\mathsf{pri})$ over parking functions. We will give a positive answer to this question by introducing two pairs of natural statistics on rooted labeled trees.

2. First involution on Dyck paths and proof of Theorem 1.1

Recall that a *Dyck path* of order n is a lattice path in $\mathbb{N}^2$ from $(0,0)$ to (n,n) using the *east step* $(1,0)$ and the *north step* $(0,1)$, which does not pass above the diagonal $y=x$. Let $\mathcal{D}_n$ be the set of all Dyck paths of order n. Define the *height of an east step* in a Dyck path to be one plus the number of north steps before this east step. For the sake of convenience, we represent a Dyck path of order n as the sequence $(h_1,h_2,\cdots,h_n)$, where h_i is the height of its i-th east step. For example, the Dyck path $(1,1,1,1,2,2,3,6,7,7,7,9,10,10)$ is drawn in the left picture of Fig. 1.

Let $\mathbf{c}=(c_1,c_2,\ldots,c_k)$ be a composition of n, i.e., all c_i's are positive integers with $c_1+c_2+\cdots+c_k=n$. A Dyck path $h_1h_2\cdots h_n\in\mathcal{D}_n$ is said to have *composition-type* $\mathbf{c}$ if

$$h_i\neq h_{i+1}\Longleftrightarrow i\in\{c_1,c_1+c_2,\ldots,c_1+c_2+\cdots+c_{k-1}\}.$$

In other words, this Dyck path begins with c_1 east steps followed by at least one north step, and then continues with c_2 east steps followed by at least one north step, and then so on. For example, the two Dyck paths in Fig. 1 both have

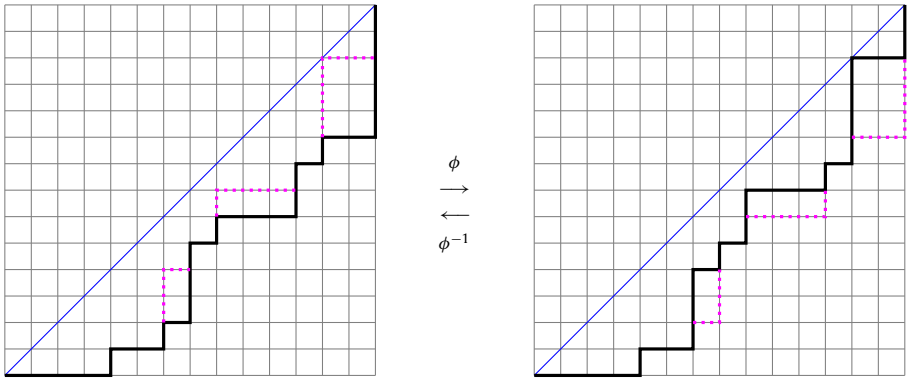


Fig. 1. An example of the construction of ϕ.

composition-type $(4, 2, 1, 1, 3, 1, 2)$. Moreover, if a Dyck path has composition-type $\mathbf{c}$ and the non-increasing rearrangement of $\mathbf{c}$ is λ, then we say this Dyck path has *partition-type* λ. Since a sequence $\alpha \in [n]^n$ is a parking function if and only if $\mathrm{inc}(\alpha) \in \mathcal{D}_n$, we have the following lemma.

Lemma 2.1. *Let $D \in \mathcal{D}_n$ be a Dyck path of partition-type $\mathbf{c} = (c_1, c_2, \ldots, c_k)$. Then the number of parking function $\alpha \in \mathbf{PF}_n$ such that $\mathrm{inc}(\alpha) = D$ equals*

$$\frac{n!}{c_1!c_2!\cdots c_k!} = \binom{n}{c_1, c_2, \ldots, c_k}.$$

Proof. Consider the many to one correspondence $\alpha \mapsto \mathrm{inc}(\alpha)$ from $\mathbf{PF}_n$ to $\mathcal{D}_n$. Then a parking function in $\mathbf{PF}_n$ that is mapped to $D = (h_1, h_2, \ldots, h_n)$ under this correspondence iff it is a rearrangement of $(h_1, h_2, \ldots, h_n)$. Since there are exactly $\frac{n!}{c_1!c_2!\cdots c_k!}$ rearrangements of $(h_1, h_2, \ldots, h_n)$, the result follows. □

For a Dyck path $D = (h_1, h_2, \ldots, h_n) \in \mathcal{D}_n$, denote $\mathsf{ret}(D)$ the number of times that D returns to the diagonal $y = x$ after the departure. Set $\mathsf{run}(D) = \mathsf{run}(h_1, h_2, \ldots, h_n)$. The following observation about the pair $(\mathsf{run}, \mathsf{pri})$ is crucial in proving Theorem 1.1.

Lemma 2.2. *For any parking function α, we have*

$$(\mathsf{run}, \mathsf{pri})(\alpha) = (\mathsf{run}, \mathsf{ret})\mathrm{inc}(\alpha),$$

considering $\mathrm{inc}(\alpha)$ *as a Dyck path.*

In view of Lemmas 2.1 and 2.2, Theorem 1.1 follows from the following involution.

Theorem 2.3. *There exists a composition-type preserving involution Φ on $\mathcal{D}_n$ that exchanges the pair of statistics* $(\mathsf{run}, \mathsf{ret})$.

Proof. Given two positive integers k and l, let

$$\mathcal{D}_n^{(k,l)} := \{D \in \mathcal{D}_n : \mathsf{run}(D) = k, \mathsf{ret}(D) = l\}. \tag{2.1}$$

For each $D \in \mathcal{D}_n^{(k,l)}$, define $\Phi(D) = \phi^{k-l}(D)$ where ϕ is a composition-type preserving invertible map defined below that decreases the statistic "run" by one but increases the statistic "ret" by one.

For a Dyck path D, the i-th maximal string of consecutive east steps is called the *i-th floor* of D. For example, the fifth floor of the Dyck path in the left picture of Fig. 1 is formed by three east steps of height 7. If $\mathsf{run}(D) = k$, then the k-th floor is called the *run floor* of D. Let us introduce two special subsets of $\mathcal{D}_n^{(k,l)}$:

- $\mathcal{A}_n^{(k,l)}$ consists of $D \in \mathcal{D}_n^{(k,l)}$ such that each floor of D before the run floor is formed by only one east step;
- $\mathcal{B}_n^{(k,l)}$ consists of $D \in \mathcal{D}_n^{(k,l)}$ such that the path D returns the diagonal only one time after the run floor.

See Fig. 2 for one example of Dyck path in $\mathcal{A}_{11}^{(3,4)}$ and one in $\mathcal{B}_{11}^{(6,4)}$ respectively.

Now we can define a bijection $\phi : \mathcal{D}_n^{(k,l)} \setminus \mathcal{A}_n^{(k,l)} \to \mathcal{D}_n^{(k-1,l+1)} \setminus \mathcal{B}_n^{(k-1,l+1)}$ that preserves the composition-type in the following algorithm. For any $D \in \mathcal{D}_n^{(k,l)} \setminus \mathcal{A}_n^{(k,l)}$, $\phi(D)$ is obtained from D by performing the following steps:

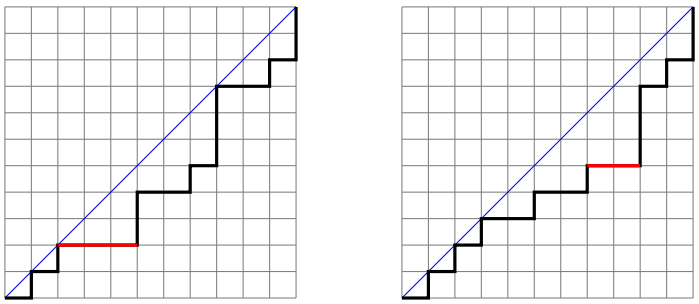


Fig. 2. An example of Dyck path in $\mathcal{A}_{11}^{(3,4)}$ and one in $\mathcal{B}_{11}^{(6,4)}$.

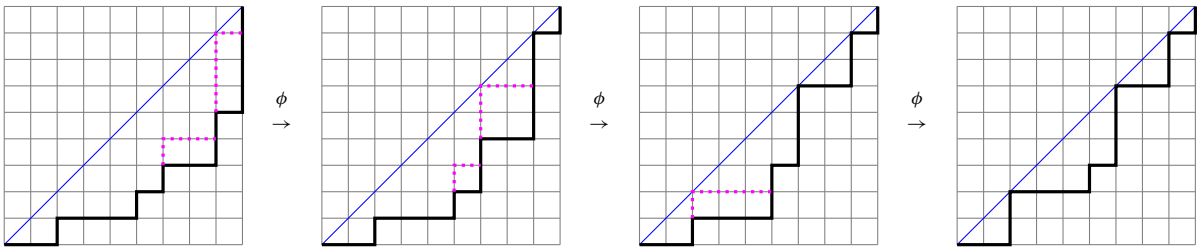


Fig. 3. An example of Φ (the pair (run, ret) is indicated under each picture).

(1) Initialization: set $i \leftarrow k$ (i is a temporary variable);
(2) increase the height of the i-th floor of D until it either touches the diagonal or meets the height of the $(i+1)$-th floor minus 1; if it touches the diagonal, then the algorithm stops and set $\phi(D)$ to be the current Dyck path; otherwise, set $i \leftarrow i+1$ and go to step (3);
(3) if the height of the i-th floor equals the height of the $(i+1)$-th floor minus 1, then set $i \leftarrow i+1$ and do (3) again; otherwise, go to (2).

It is clear that the above algorithm will output a Dyck path $\phi(D) \in \mathcal{D}_n^{(k-1,l+1)} \setminus \mathcal{B}_n^{(k-1,l+1)}$. See Fig. 1 for an example of ϕ.

To see that ϕ is a bijection, we define its inverse ϕ^{-1} explicitly. For $D \in \mathcal{D}_n^{(k-1,l+1)} \setminus \mathcal{B}_n^{(k-1,l+1)}$, suppose that the j-th floor of D is the first floor that touches the diagonal after the run floor. We perform the following steps to retrieve $\phi^{-1}(D)$ from D:

(1) Initialization: set $i \leftarrow j$;
(2) decrease the height of the i-th floor of D until it meets the height of the $(i-1)$-th floor plus 1; if the $(i-1)$-th floor is the run floor, then the algorithm stops and set $\phi^{-1}(D)$ to be the current Dyck path; otherwise, set $i \leftarrow i-1$ and go to step (3);
(3) if the height of the i-th floor equals the height of the $(i-1)$-th floor plus 1, then set $i \leftarrow i-1$ and do (3) again; otherwise, go to (2).

It is routine to check that ϕ^{-1} and ϕ are inverse to each other and preserve the composition-type of Dyck paths, as desired.

Finally, we need to prove that Φ is well-defined. Note that when $k > l$, then there are at least $k-l$ floors between the first big floor and the run floor (including run floor), where a *big floor* is a floor formed by at least two east steps. As the algorithm ϕ increases exactly one of such a floor (starting from the run floor) each time, Φ is well-defined when $k > l$. See Fig. 3 for an example of Φ. The case for $k < l$ is similar and so Φ is actually an involution. This completes the proof of the theorem. □

In the above proof, since $\mathcal{A}_n^{(k,l)}$ (resp. $\mathcal{B}_n^{(k,l)}$) is nonempty if and only if $k \leqslant l$ (resp. $k \geqslant l$), we actually prove the following inequality.

Corollary 2.4. *For $k \geqslant l+2$, we have $|\mathcal{D}_n^{(k,l)}| < |\mathcal{D}_n^{(k-1,l+1)}|$, where $\mathcal{D}_n^{(k,l)}$ is defined by (2.1). Consequently, $|\mathbf{PF}_n^{(k,l)}| < |\mathbf{PF}_n^{(k-1,l+1)}|$ for $k \geqslant l+2$, where*

$$\mathbf{PF}_n^{(k,l)} := \{\alpha \in \mathbf{PF}_n : \mathsf{run}(\alpha) = k, \mathsf{pri}(\alpha) = l\}.$$

Some values of $|\mathcal{D}_n^{(k,l)}|$ and $|\mathbf{PF}_n^{(k,l)}|$ are listed as follows:

$$\{|\mathcal{D}_6^{(k,l)}|\} = \begin{bmatrix} 14 & 14 & 9 & 4 & 1 & 0 \\ 14 & 14 & 9 & 4 & 1 & 0 \\ 9 & 9 & 6 & 3 & 1 & 0 \\ 4 & 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \{|\mathcal{D}_7^{(k,l)}|\} = \begin{bmatrix} 42 & 42 & 28 & 14 & 5 & 1 & 0 \\ 42 & 42 & 28 & 14 & 5 & 1 & 0 \\ 28 & 28 & 19 & 10 & 4 & 1 & 0 \\ 14 & 14 & 10 & 6 & 3 & 1 & 0 \\ 5 & 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and

$$\{|\mathbf{PF}_5^{(k,l)}|\} = \begin{bmatrix} 41 & 75 & 80 & 60 & 0 \\ 75 & 105 & 110 & 60 & 0 \\ 80 & 110 & 80 & 60 & 0 \\ 60 & 60 & 60 & 60 & 0 \\ 0 & 0 & 0 & 0 & 120 \end{bmatrix}, \quad \{|\mathbf{PF}_6^{(k,l)}|\} = \begin{bmatrix} 459 & 806 & 840 & 660 & 360 & 0 \\ 806 & 1086 & 1110 & 840 & 360 & 0 \\ 840 & 1110 & 990 & 660 & 360 & 0 \\ 660 & 840 & 660 & 480 & 360 & 0 \\ 360 & 360 & 360 & 360 & 360 & 0 \\ 0 & 0 & 0 & 0 & 0 & 720 \end{bmatrix}.$$

3. On three bijections between Dyck paths and plane trees

We will investigate three bijections between Dyck paths and plane trees, two of which are known and the third one is based on the level order traversal on plane trees.

3.1. Second involution on Dyck paths via plane trees

In this section, we construct another simple involution on Dyck paths via plane trees that also proves Theorem 1.1, in view of Lemmas 2.1 and 2.2.

Theorem 3.1. *There exists a partition-type preserving involution Ψ on $\mathcal{D}_n$ that exchanges the pair of statistics* (run, ret).

Recall that a plane tree is a rooted tree in which the children of each node are linearly ordered. Denote by $\mathcal{T}_n$ the set of all plane trees with n edges. For $T \in \mathcal{T}_n$, let $\psi(T)$ be the mirror symmetry of T (see Fig. 5). We consider the following three statistics on T:

- the number of branches of T at the root, denoted $\mathsf{bran}(T)$;
- the number of non-root nodes visited before backtracking when the tree T is traversed in preorder, denoted $\mathsf{larm}(T)$;
- the number of non-root nodes visited before backtracking when the tree $\psi(T)$ is traversed in preorder, denoted $\mathsf{rarm}(T)$. Note that $\mathsf{rarm}(T) = \mathsf{larm}(\psi(T))$.

For example, for the tree in the second picture of Fig. 5, we have $\mathsf{bran}(T) = 3$, $\mathsf{larm}(T) = 1$ and $\mathsf{rarm}(T) = 2$.

For a plane tree T, the degree of a node v, denoted deg_v, in T is the number of its children. If v_1 is the root of T and $v_2, v_3, \ldots, v_k$ are the interior vertices of T listed in preorder, then we say T has composition-type $(\mathsf{deg}_{v_1}, \mathsf{deg}_{v_2}, \ldots, \mathsf{deg}_{v_k})$. For example, the tree in the second picture of Fig. 5 has composition-type $(3, 2, 1, 1)$.

There are two well-known bijections (see [2,5]) between plane trees and Dyck paths which transform the above three tree statistics to Dyck paths.

- The first bijection τ_1 maps a plane tree to a Dyck path by recording the steps when the tree is traversed in preorder: we record an east step when we go down an edge and a north step when going up. See Fig. 4 for an example of τ_1.
- The second bijection τ_2 maps a plane tree to a Dyck path by recording the steps when the tree is traversed in preorder: whenever we visit a node of degree i (for the first time), except the last leaf, we record i east steps followed by one north step. See Fig. 5 for two examples of τ_2.

For $D \in \mathcal{D}_n$, denote by $\mathsf{iest}(D)$ (resp. $\mathsf{lnor}(D)$) the number of initial consecutive east (resp. last consecutive north) steps of D. The bijections τ_1 and τ_2 have the following properties, which are clear from the above constructions.

Lemma 3.2. *For any plane tree T, we have*

$$(\mathsf{bran}, \mathsf{larm}, \mathsf{rarm})(T) = (\mathsf{ret}, \mathsf{iest}, \mathsf{lnor})\tau_1(T) \tag{3.1}$$

and

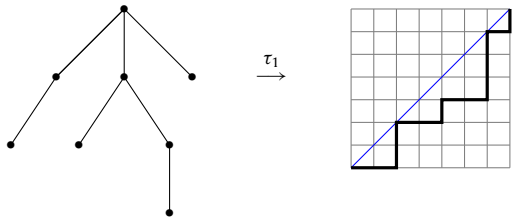


Fig. 4. An example of τ_1.

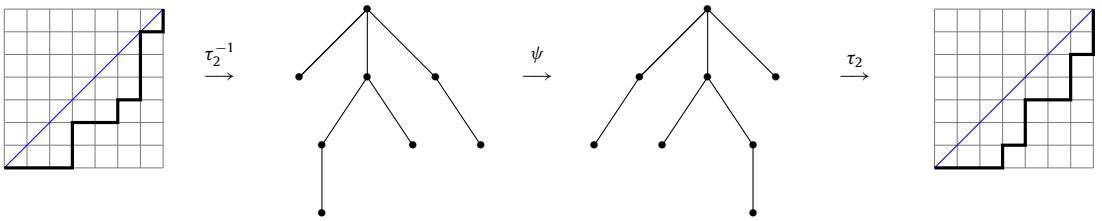


Fig. 5. An example of Ψ.

$$(\mathsf{bran}, \mathsf{larm}, \mathsf{rarm})(T) = (\mathsf{iest}, \mathsf{run}, \mathsf{ret})\tau_2(T). \tag{3.2}$$

Moreover, the bijection τ_2 is composition-type preserving. Consequently, we have the equidistribution over Dyck paths

$$(\mathsf{ret}, \mathsf{iest}, \mathsf{lnor}) \sim (\mathsf{iest}, \mathsf{run}, \mathsf{ret}).$$

Proof. We will only prove that $\mathsf{rarm}(T) = \mathsf{ret}(\tau_2(T))$, as the other correspondences are more apparent. A plane tree whose right arm has length 1 is called a *prime plane tree*, while a Dyck path that returns to the diagonal only at last step is named a *prime Dyck path*. As every plane tree can be decomposed along the right arm into a sequence of prime plane trees, we only need to show that $\tau_2(T)$ is a prime Dyck path for a prime plane tree T.

Let r be the root of T and u be the rightmost child of r. For any node v other than r and u, denote D_v the partial Dyck path recorded in the algorithm τ_2 before visiting v. Note that the node r (resp. u or v) contributes one north (resp. east) step to D_v, each node other than r before v (under the preorder) contributes both one east step and one north step to D_v, and each node other than u after v (under the preorder) may contribute one east step to D_v (only if it is a sibling of v). Therefore, D_v contains strictly more east steps than north steps, which asserts that $\tau_2(T)$ is a prime Dyck path. □

We can now construct the bijection Ψ for Theorem 3.1.

Proof of Theorem 3.1. Set $\Psi = \tau_2 \circ \psi \circ \tau_2^{-1}$. The required properties of Ψ then follow from (3.2) and the fact that τ_2 is composition-type preserving. □

3.2. Two symmetric pairs on rooted labeled trees

In this section, we provide two pairs of statistics over rooted labeled trees that have the same distribution as $(\mathsf{run}, \mathsf{pri})$ over parking functions.

A rooted labeled tree on $[n]_0 := \{0, 1, 2, \ldots, n\}$ is a plane tree labeled by $[n]_0$ satisfying

- the root is labeled 0;
- the labels of children of any node are increasing from left to right.

See Fig. 6 for a rooted labeled tree on $[7]_0$. The set of all rooted labeled trees on $[n]_0$ is denoted by $\mathbf{LT}_n$. It is well known [9] that $|\mathbf{LT}_n| = (n+1)^{n-1} = |\mathbf{PF}_n|$. Since the labeling is irrelevant for the three statistics $(\mathsf{bran}, \mathsf{larm}, \mathsf{rarm})$, they can be extended naturally to rooted labeled trees.

The following lemma is obvious from the definition of rooted labeled trees.

Lemma 3.3. *For a plane tree $T \in \mathcal{T}_n$ with composition-type $(c_1, c_2, \ldots, c_k)$, the number of rooted labeled trees in $\mathbf{LT}_n$ whose underlying plane tree is T equals*

$$\frac{n!}{c_1!c_2!\cdots c_k!} = \binom{n}{c_1, c_2, \ldots, c_k}.$$

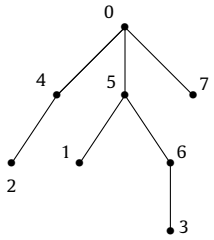


Fig. 6. A rooted labeled tree on $[7]_0$.

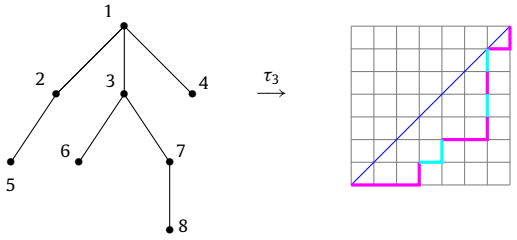


Fig. 7. The level order on a tree and an example of τ_3.

It then follows from (3.2) and the fact that τ_2 is composition-type preserving that

Proposition 3.4. *The pair* (larm, rarm) *over* $\mathbf{LT}_n$ *has the same distribution as the pair* (run, pri) *over* $\mathbf{PF}_n$.

Remark 3.5. The result that larm over rooted labeled trees has the same distribution as run over parking functions was proved in [4].

Now we introduce the third bijection between plane trees and Dyck paths. For a plane tree, the root is at level 0 and any node that is a child of a node at level i is said to be at level $i+1$ for any i. A *level order traversal* is a traversal which starting from the root and traversing the nodes at level 1 from left to right, and then level 2, and so on.

In the same spirit as τ_2, the third bijection τ_3 maps a plane tree to a Dyck path by recording the steps when the tree is traversed in level order: whenever we visit a node of degree i (for the first time), except the last leaf, we record i east steps followed by one north step. See Fig. 7 for an example of τ_3.

For $T \in \mathcal{T}_n$, introduce two tree statistics based on the levels of T:

- the number of nodes visited before the first leaf when the tree T is traversed in level order, denoted $\widetilde{\mathsf{larm}}(T)$;
- the number of saturated levels (level 0 excluded) in T, denoted $\mathsf{satu}(T)$, where a level is said to be *saturated* if all nodes in this level other than the rightmost one are leaves.

Note that $\mathsf{larm}(T)$ can also be interpreted as the number of nodes visited before the first leaf when the tree T is traversed in preorder, that is the reason we use the notation $\widetilde{\mathsf{larm}}(T)$ above. The following features of τ_3 can be verified from its construction easily.

Lemma 3.6. *For any plane tree T, we have*

$$(\mathsf{bran}, \widetilde{\mathsf{larm}}, \mathsf{satu})(T) = (\mathsf{iest}, \mathsf{run}, \mathsf{ret})\tau_3(T). \tag{3.3}$$

Thus, we have the equidistribution over plane trees

$$(\mathsf{bran}, \widetilde{\mathsf{larm}}, \mathsf{satu}) \sim (\mathsf{bran}, \mathsf{larm}, \mathsf{rarm}).$$

The pair $(\widetilde{\mathsf{larm}}, \mathsf{satu})$ is extended naturally to rooted labeled tree. As a consequence of τ_3 and (3.3), we have

Proposition 3.7. *The pair* $(\widetilde{\mathsf{larm}}, \mathsf{satu})$ *over* $\mathbf{LT}_n$ *has the same distribution as the pair* (run, pri) *over* $\mathbf{PF}_n$.

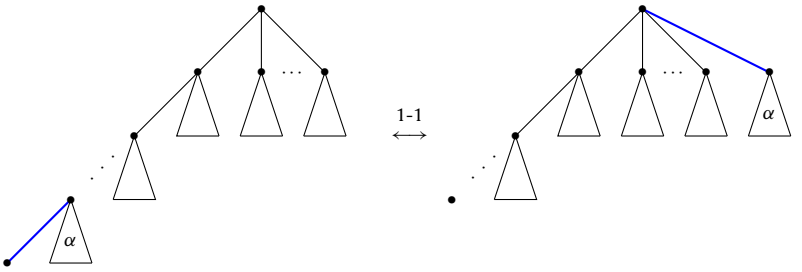


Fig. 8. The illustration of the one-to-one correspondence $T \mapsto T'$.

3.3. Catalan's triangle and symmetric pairs

The *Catalan's triangle* (also known as *ballot numbers*) $C_{n,k}$ can be defined by

$$C_{n,k} = \frac{k}{2n-k}\binom{2n-k}{n}$$

for $n \geqslant k \geqslant 1$. Deutsch [2] proved that

$$|\{D \in \mathcal{D}_n : \mathsf{iest}(D) = k\}| = C_{n,k}.$$

The reader is referred to [1] for more combinatorial interpretations of Catalan's triangle. The following result provides more interpretations of Catalan's triangle, some of which may not be new.

Proposition 3.8. *We have the following three classes of equidistributed symmetric pairs.*

(1) Any pair in $\{(\mathsf{larm}, \mathsf{bran}), (\mathsf{rarm}, \mathsf{bran}), (\widetilde{\mathsf{larm}}, \mathsf{bran})\}$ *over* $\mathcal{T}_n$ *is symmetric and has the same distribution as any pair in* $\{(\mathsf{run}, \mathsf{iest}), (\mathsf{ret}, \mathsf{iest}), (\mathsf{ret}, \widetilde{\mathsf{lnor}})\}$ *over* $\mathcal{D}_n$.
(2) Any pair in $\{(\mathsf{larm}, \mathsf{rarm}), (\widetilde{\mathsf{larm}}, \mathsf{satu})\}$ *over* $\mathcal{T}_n$ *is symmetric and has the same distribution as any pair in* $\{(\mathsf{run}, \mathsf{ret}), (\mathsf{iest}, \mathsf{lnor})\}$ *over* $\mathcal{D}_n$.
(3) The pair $(\mathsf{run}, \mathsf{lnor})$ *is symmetric over* $\mathcal{D}_n$.

In particular, any statistic in $\{\mathsf{bran}, \mathsf{larm}, \mathsf{rarm}, \widetilde{\mathsf{larm}}, \mathsf{satu}\}$ *over plane trees or any statistic in* $\{\mathsf{run}, \mathsf{ret}, \mathsf{iest}, \mathsf{lnor}\}$ *over Dyck paths interprets Catalan's triangle.*

Proof. Because of Lemmas 3.2 and 3.6, we only need to prove that the pair $(\mathsf{larm}, \mathsf{bran})$ is symmetric over $\mathcal{T}_n$ and the pair $(\mathsf{run}, \mathsf{lnor})$ is symmetric over $\mathcal{D}_n$.

First we prove that the pair $(\mathsf{larm}, \mathsf{bran})$ is symmetric over $\mathcal{T}_n$. Actually, we have the stronger symmetry $|\mathcal{T}_n^{(k,l)}| = |\mathcal{T}_n^{(k-1,l+1)}|$ whenever $k \geqslant 2$ and $l \geqslant 1$, where

$$\mathcal{T}_n^{(k,l)} := \{T \in \mathcal{T}_n : \mathsf{larm}(T) = k, \mathsf{bran}(T) = l\}.$$

This is a consequence of the one-to-one correspondence $T \mapsto T'$ between $\mathcal{T}_n^{(k,l)}$ and $\mathcal{T}_n^{(k-1,l+1)}$, where T' is obtained from T by detaching the subtree rooted at the last-but-one node on the left arm and attaching to the root as rightmost subtree (see Fig. 8 for the illustration).

It remains to show that $(\mathsf{run}, \mathsf{lnor})$ is symmetric over $\mathcal{D}_n$. We will do this by using generating functions. Let

$$C(t,p,q) := \sum_{n\geqslant 1} t^n \sum_{D\in\mathcal{D}_n} p^{\mathsf{run}(D)} q^{\mathsf{lnor}(D)} = tpq + (p^2q + pq^2)t^2 + (pq^3 + p^3q + 2p^2q^2 + pq)t^3 + \cdots.$$

We will encode a Dyck path by a word over $\{E, N\}$, where E is an east step and N is a north step. We decompose a Dyck path D according to the following four cases:

i) $D = EN$. This case contributes the generating function tpq.
ii) $D = END'$, where D' is a Dyck path. This case contributes the generating function $tpC(t,p,q)$.
iii) $D = ED'N$, where D' is a Dyck path. This case contributes the generating function $tqC(t,p,q)$.
iv) $D = ED_1ND_2$, where D_1 and D_2 are two Dyck paths. This case contributes the generating function $tC(t,p,1)C(t,1,q)$.

Summing over all the above cases results in the functional equation for $C(t,p,q)$:

$$C(t,p,q) = tpq + t(p+q)C(t,p,q) + tC(t,p,1)C(t,1,q),$$

which is symmetric in p and q. This proves the symmetry of $(\mathsf{run}, \mathsf{lnor})$ over Dyck paths. □

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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